```python
import pandas as pd
import numpy as np
from sklearn.preprocessing import MinMaxScaler
# Load the dataset
file_path = r"C:\Users\Dell\Documents\Modified_Crime_Dataset.csv"
data = pd.read_csv(file_path)

# Display basic info about the dataset
print("Dataset Info:")
print(data.info())
print("\nFirst Few Rows:")
print(data.head())

# 1. Data Transformation: Normalize Numerical Columns
numerical_cols = data.select_dtypes(include=['float64', 'int64']).columns
scaler = MinMaxScaler()
data[numerical_cols] = scaler.fit_transform(data[numerical_cols])
print("\nNormalized Data (First Few Rows):")
print(data[numerical_cols].head())

# 2. Encoding Categorical Data
# One-hot encode STATE/UT and DISTRICT
categorical_cols = ['STATE/UT', 'DISTRICT']
data_encoded = pd.get_dummies(data, columns=categorical_cols,
         drop_first=True)
print("\nEncoded Data (First Few Rows):")
print(data_encoded.head())

# 3. Aggregation and Grouping: Summarize by YEAR and STATE/UT
crime_summary_by_year_state = data.groupby(['YEAR',
         'STATE/UT']).sum().reset_index()
print("\nCrime Summary by Year and State (First Few Rows):")
print(crime_summary_by_year_state.head())

# 4. Interaction Features: Combine related features
if 'RAPE' in data.columns and 'OTHER RAPE' in data.columns:
    data_encoded['TOTAL_RAPE'] = data['RAPE'] + data['OTHER RAPE']
    print("\nInteraction Feature - TOTAL_RAPE (First Few Rows):")
    print(data_encoded[['RAPE', 'OTHER RAPE', 'TOTAL_RAPE']].head())
else:
    print("Columns 'RAPE' or 'OTHER RAPE' not found, skipping interaction
         feature creation.")

# 5. Binning: Group TOTAL IPC CRIMES into severity levels
if 'TOTAL IPC CRIMES' in data.columns:
    bins = [0, 10000, 50000, 100000, np.inf]
    labels = ['Low', 'Moderate', 'High', 'Very High']
    data_encoded['CRIME_SEVERITY'] = pd.cut(data['TOTAL IPC CRIMES'],
         bins=bins, labels=labels)
```

```
    print("\nBinned Data - CRIME_SEVERITY (First Few Rows):")
    print(data_encoded[['TOTAL IPC CRIMES', 'CRIME_SEVERITY']].head())
else:
    print("Column 'TOTAL IPC CRIMES' not found, skipping binning.")

# Save the processed dataset
output_path = r"C:\Users\Dell\Documents\Processed Modified_Crime_Dataset.csv"
data_encoded.to_csv(output_path, index=False)
print(f"\nProcessed dataset saved to: {output_path}")
```

```
Dataset Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9017 entries, 0 to 9016
Columns: 871 entries, YEAR to TOTAL_RAPE
dtypes: bool(841), float64(30)
memory usage: 9.3 MB
None

First Few Rows:
   YEAR    MURDER  ATTEMPT TO MURDER  \
0   0.0  0.013288           0.007534
1   0.0  0.019866           0.015696
2   0.0  0.013288           0.007157
3   0.0  0.010525           0.006655
4   0.0  0.010788           0.008413

   CULPABLE HOMICIDE NOT AMOUNTING TO MURDER      RAPE  CUSTODIAL RAPE  \
0                                   0.010520  0.014599             0.0
1                                   0.000619  0.006715             0.0
2                                   0.001238  0.007883             0.0
3                                   0.000619  0.005839             0.0
4                                   0.000619  0.006715             0.0

   OTHER RAPE  KIDNAPPING & ABDUCTION  \
0    0.014599                0.005181
1    0.006715                0.005970
2    0.007883                0.006646
3    0.005839                0.002816
4    0.006715                0.005519

   KIDNAPPING AND ABDUCTION OF WOMEN AND GIRLS  \
0                                     0.003793
1                                     0.003793
2                                     0.004298
3                                     0.002528
4                                     0.003287

   KIDNAPPING AND ABDUCTION OF OTHERS  ...  DISTRICT_WASHIM  \
0                            0.006623  ...            False
1                            0.009520  ...            False
2                            0.010348  ...            False
3                            0.002070  ...            False
4                            0.009520  ...            False
```

```
   DISTRICT_WAYANADU  DISTRICT_WEST  DISTRICT_WEST GODAVARI  DISTRICT_WOKHA  
\
0              False          False                   False           False  
1              False          False                   False           False  
2              False          False                   False           False  
3              False          False                   False           False  
4              False          False                   False           False  

   DISTRICT_YADGIRI  DISTRICT_YAMUNANAGAR  DISTRICT_YAVATMAL  \
0             False                 False              False   
1             False                 False              False   
2             False                 False              False   
3             False                 False              False   
4             False                 False              False   

   DISTRICT_ZUNHEBOTO  TOTAL_RAPE  
0               False    0.029197  
1               False    0.013431  
2               False    0.015766  
3               False    0.011679  
4               False    0.013431  

[5 rows x 871 columns]

Normalized Data (First Few Rows):
   YEAR    MURDER  ATTEMPT TO MURDER  \
0   0.0  0.013288           0.007534   
1   0.0  0.019866           0.015696   
2   0.0  0.013288           0.007157   
3   0.0  0.010525           0.006655   
4   0.0  0.010788           0.008413   

   CULPABLE HOMICIDE NOT AMOUNTING TO MURDER      RAPE  CUSTODIAL RAPE  \
0                                   0.010520  0.014599             0.0   
1                                   0.000619  0.006715             0.0   
2                                   0.001238  0.007883             0.0   
3                                   0.000619  0.005839             0.0   
4                                   0.000619  0.006715             0.0   

   OTHER RAPE  KIDNAPPING & ABDUCTION  \
0    0.014599                0.005181   
1    0.006715                0.005970   
2    0.007883                0.006646   
3    0.005839                0.002816   
4    0.006715                0.005519   

   KIDNAPPING AND ABDUCTION OF WOMEN AND GIRLS  \
0                                     0.003793   
1                                     0.003793   
2                                     0.004298   
3                                     0.002528   
4                                     0.003287   

   KIDNAPPING AND ABDUCTION OF OTHERS  ...  COUNTERFIETING     ARSON  \
```

```
0                          0.006623  ...        0.001669  0.010601
1                          0.009520  ...        0.013356  0.024382
2                          0.010348  ...        0.015025  0.013428
3                          0.002070  ...        0.003339  0.008127
4                          0.009520  ...        0.005008  0.014488

   HURT/GREVIOUS HURT  DOWRY DEATHS  \
0            0.019923      0.006891
1            0.027181      0.003015
2            0.036781      0.006029
3            0.014004      0.007321
4            0.021914      0.005168

   ASSAULT ON WOMEN WITH INTENT TO OUTRAGE HER MODESTY  \
0                                           0.020933
1                                           0.016578
2                                           0.015735
3                                           0.017702
4                                           0.015313

   INSULT TO MODESTY OF WOMEN  CRUELTY BY HUSBAND OR HIS RELATIVES  \
0                    0.006841                             0.008809
1                    0.004829                             0.007752
2                    0.016700                             0.009363
3                    0.007646                             0.002869
4                    0.011670                             0.012434

   IMPORTATION OF GIRLS FROM FOREIGN COUNTRIES  CAUSING DEATH BY NEGLIGENCE
\
0                                          0.0                     0.011259
1                                          0.0                     0.016795
2                                          0.0                     0.025131
3                                          0.0                     0.014494
4                                          0.0                     0.026810

   TOTAL_RAPE
0    0.014599
1    0.006715
2    0.007883
3    0.005839
4    0.006715

[5 rows x 30 columns]

---------------------------------------------------------------------------
KeyError                                  Traceback (most recent call last)
Cell In[19], line 24
     21 # 2. Encoding Categorical Data
     22 # One-hot encode STATE/UT and DISTRICT
     23 categorical_cols = ['STATE/UT', 'DISTRICT']
---> 24 data_encoded = pd.get_dummies(data, columns=categorical_cols,
drop_first=True)
     25 print("\nEncoded Data (First Few Rows):")
     26 print(data_encoded.head())
```

```
File ~\anaconda3\Lib\site-packages\pandas\core\reshape\encoding.py:169, in
get_dummies(data, prefix, prefix_sep, dummy_na, columns, sparse, drop_first,
dtype)
    167     raise TypeError("Input must be a list-like for parameter
`columns`")
    168 else:
--> 169     data_to_encode = data[columns]
    171 # validate prefixes and separator to avoid silently dropping cols
    172 def check_len(item, name: str):

File ~\anaconda3\Lib\site-packages\pandas\core\frame.py:4108, in
DataFrame.__getitem__(self, key)
   4106     if is_iterator(key):
   4107         key = list(key)
-> 4108     indexer = self.columns._get_indexer_strict(key, "columns")[1]
   4110 # take() does not accept boolean indexers
   4111 if getattr(indexer, "dtype", None) == bool:

File ~\anaconda3\Lib\site-packages\pandas\core\indexes\base.py:6200, in
Index._get_indexer_strict(self, key, axis_name)
   6197 else:
   6198     keyarr, indexer, new_indexer = self._reindex_non_unique(keyarr)
-> 6200 self._raise_if_missing(keyarr, indexer, axis_name)
   6202 keyarr = self.take(indexer)
   6203 if isinstance(key, Index):
   6204     # GH 42790 - Preserve name from an Index

File ~\anaconda3\Lib\site-packages\pandas\core\indexes\base.py:6249, in
Index._raise_if_missing(self, key, indexer, axis_name)
   6247 if nmissing:
   6248     if nmissing == len(indexer):
-> 6249         raise KeyError(f"None of [{key}] are in the [{axis_name}]")
   6251     not_found = list(ensure_index(key)[missing_mask.nonzero()
[0]].unique())
   6252     raise KeyError(f"{not_found} not in index")

KeyError: "None of [Index(['STATE/UT', 'DISTRICT'], dtype='object')] are in
the [columns]"
```

```python
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.feature_selection import VarianceThreshold, SelectKBest, chi2,
         mutual_info_regression
from sklearn.preprocessing import MinMaxScaler

# ============================
# Load Dataset
# ============================
file_path = r"C:\Users\Dell\Documents\Modified_Crime_Dataset.csv"
data = pd.read_csv(file_path)
```

```python
# Recreate 'Total_Crimes' column if it doesn't exist
if 'Total_Crimes' not in data.columns:
    crime_columns = data.columns[3:]  # Adjust based on your dataset
    data['Total_Crimes'] = data[crime_columns].sum(axis=1)

# Define features (X) and target variable (y)
X = data.drop(columns=['Total_Crimes'])  # Features
y = data['Total_Crimes']  # Target variable

# Ensure all features are numeric
X = pd.get_dummies(X, drop_first=True)

# 1. Variance Threshold
# =============================
from sklearn.feature_selection import VarianceThreshold
from sklearn.preprocessing import StandardScaler

# Check variance of all features before applying threshold
feature_variances = X.var()
print("\nFeature Variances (Before Scaling):")
print(feature_variances)

# Lower the threshold for variance
threshold = 0.01  # Reduced variance threshold

# Standardize features for more meaningful variance calculations
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# Apply VarianceThreshold
selector = VarianceThreshold(threshold)
X_high_variance = selector.fit_transform(X_scaled)

# Extract selected features
selected_features_var = X.columns[selector.get_support()].tolist()
print("\nSelected Features using Variance Threshold (After Scaling):",
         selected_features_var)

# If no features are selected, check for features with low variance for
         debugging
if not selected_features_var:
    low_variance_features = feature_variances[feature_variances < threshold]
    print("\nFeatures with Low Variance:")
    print(low_variance_features)
```

```
Feature Variances (Before Scaling):
YEAR                                          0.099146
MURDER                                        0.001854
ATTEMPT TO MURDER                             0.001452
CULPABLE HOMICIDE NOT AMOUNTING TO MURDER     0.001357
RAPE                                          0.003101
                                                ...
```

```
DISTRICT_YADGIRI                              0.000333
DISTRICT_YAMUNANAGAR                          0.001329
DISTRICT_YAVATMAL                             0.001329
DISTRICT_ZUNHEBOTO                            0.001329
TOTAL_RAPE                                    0.012405
Length: 871, dtype: float64

Selected Features using Variance Threshold (After Scaling): ['YEAR',
'MURDER', 'ATTEMPT TO MURDER', 'CULPABLE HOMICIDE NOT AMOUNTING TO MURDER',
'RAPE', 'CUSTODIAL RAPE', 'OTHER RAPE', 'KIDNAPPING & ABDUCTION', 'KIDNAPPING
AND ABDUCTION OF WOMEN AND GIRLS', 'KIDNAPPING AND ABDUCTION OF OTHERS',
'DACOITY', 'PREPARATION AND ASSEMBLY FOR DACOITY', 'ROBBERY', 'BURGLARY',
'THEFT', 'AUTO THEFT', 'OTHER THEFT', 'RIOTS', 'CRIMINAL BREACH OF TRUST',
'CHEATING', 'COUNTERFIETING', 'ARSON', 'HURT/GREVIOUS HURT', 'DOWRY DEATHS',
'ASSAULT ON WOMEN WITH INTENT TO OUTRAGE HER MODESTY', 'INSULT TO MODESTY OF
WOMEN', 'CRUELTY BY HUSBAND OR HIS RELATIVES', 'IMPORTATION OF GIRLS FROM
FOREIGN COUNTRIES', 'CAUSING DEATH BY NEGLIGENCE', 'STATE/UT_ANDHRA PRADESH',
'STATE/UT_ARUNACHAL PRADESH', 'STATE/UT_ASSAM', 'STATE/UT_BIHAR',
'STATE/UT_CHANDIGARH', 'STATE/UT_CHHATTISGARH', 'STATE/UT_D & N HAVELI',
'STATE/UT_DAMAN & DIU', 'STATE/UT_DELHI UT', 'STATE/UT_GOA',
'STATE/UT_GUJARAT', 'STATE/UT_HARYANA', 'STATE/UT_HIMACHAL PRADESH',
'STATE/UT_JAMMU & KASHMIR', 'STATE/UT_JHARKHAND', 'STATE/UT_KARNATAKA',
'STATE/UT_KERALA', 'STATE/UT_LAKSHADWEEP', 'STATE/UT_MADHYA PRADESH',
'STATE/UT_MAHARASHTRA', 'STATE/UT_MANIPUR', 'STATE/UT_MEGHALAYA',
'STATE/UT_MIZORAM', 'STATE/UT_NAGALAND', 'STATE/UT_ODISHA',
'STATE/UT_PUDUCHERRY', 'STATE/UT_PUNJAB', 'STATE/UT_RAJASTHAN',
'STATE/UT_SIKKIM', 'STATE/UT_TAMIL NADU', 'STATE/UT_TRIPURA', 'STATE/UT_UTTAR
PRADESH', 'STATE/UT_UTTARAKHAND', 'STATE/UT_WEST BENGAL', 'DISTRICT_24
PARGANAS SOUTH', 'DISTRICT_A and N ISLANDS', 'DISTRICT_ADILABAD',
'DISTRICT_AGRA', 'DISTRICT_AHMEDABAD COMMR.', 'DISTRICT_AHMEDABAD RURAL',
'DISTRICT_AHMEDNAGAR', 'DISTRICT_AHWA-DANG', 'DISTRICT_AIZAWL',
'DISTRICT_AJMER', 'DISTRICT_AKOLA', 'DISTRICT_ALAPUZHA', 'DISTRICT_ALIGARH',
'DISTRICT_ALIRAJPUR', 'DISTRICT_ALLAHABAD', 'DISTRICT_ALMORA',
'DISTRICT_ALWAR', 'DISTRICT_AMBALA', 'DISTRICT_AMBALA RURAL',
'DISTRICT_AMBALA URBAN', 'DISTRICT_AMBEDKAR NAGAR', 'DISTRICT_AMRAVATI
COMMR.', 'DISTRICT_AMRAVATI RURAL', 'DISTRICT_AMRELI', 'DISTRICT_AMRITSAR',
'DISTRICT_AMRITSAR RURAL', 'DISTRICT_ANAND', 'DISTRICT_ANANTAPUR',
'DISTRICT_ANANTNAG', 'DISTRICT_ANDAMAN', 'DISTRICT_ANGUL', 'DISTRICT_ANJAW',
'DISTRICT_ANUPPUR', 'DISTRICT_ARARIA', 'DISTRICT_ARIYALUR', 'DISTRICT_ARWAL',
'DISTRICT_ASANSOL', 'DISTRICT_ASHOK NAGAR', 'DISTRICT_AURAIYA',
'DISTRICT_AURANGABAD', 'DISTRICT_AURANGABAD COMMR.', 'DISTRICT_AURANGABAD
RURAL', 'DISTRICT_AWANTIPORA', 'DISTRICT_AZAMGARH', 'DISTRICT_BADAUN',
'DISTRICT_BADDIPOLICEDIST', 'DISTRICT_BAGAHA', 'DISTRICT_BAGALKOT',
'DISTRICT_BAGESHWAR', 'DISTRICT_BAGHPAT', 'DISTRICT_BAHRAICH',
'DISTRICT_BAKSA', 'DISTRICT_BALAGHAT', 'DISTRICT_BALASORE',
'DISTRICT_BALLIA', 'DISTRICT_BALOD', 'DISTRICT_BALODA BAZAR',
'DISTRICT_BALRAMPUR', 'DISTRICT_BANDA', 'DISTRICT_BANDIPORA',
'DISTRICT_BANGALORE COMMR.', 'DISTRICT_BANGALORE RURAL', 'DISTRICT_BANKA',
'DISTRICT_BANKURA', 'DISTRICT_BANSWARA', 'DISTRICT_BARABANKI',
'DISTRICT_BARAGARH', 'DISTRICT_BARAMULLA', 'DISTRICT_BARAN',
'DISTRICT_BAREILLY', 'DISTRICT_BARMER', 'DISTRICT_BARNALA',
'DISTRICT_BARPETA', 'DISTRICT_BARWANI', 'DISTRICT_BASKA', 'DISTRICT_BASTI',
'DISTRICT_BATALA', 'DISTRICT_BDN CP', 'DISTRICT_BEED', 'DISTRICT_BEGUSARAI',
'DISTRICT_BELGAUM', 'DISTRICT_BELLARY', 'DISTRICT_BEMETARA',
```

'DISTRICT_BERHAMPUR', 'DISTRICT_BETTIAH', 'DISTRICT_BETUL',
'DISTRICT_BHABHUA', 'DISTRICT_BHADRAK', 'DISTRICT_BHAGALPUR',
'DISTRICT_BHANDARA', 'DISTRICT_BHARATPUR', 'DISTRICT_BHARUCH',
'DISTRICT_BHATINDA', 'DISTRICT_BHAVNAGAR', 'DISTRICT_BHILWARA',
'DISTRICT_BHIM NAGAR', 'DISTRICT_BHIND', 'DISTRICT_BHIWANI',
'DISTRICT_BHOJPUR', 'DISTRICT_BHOPAL', 'DISTRICT_BHOPAL RLY.',
'DISTRICT_BIDAR', 'DISTRICT_BIEO', 'DISTRICT_BIJAPUR', 'DISTRICT_BIJNOR',
'DISTRICT_BIKANER', 'DISTRICT_BILASPUR', 'DISTRICT_BIRBHUM',
'DISTRICT_BISHNUPUR', 'DISTRICT_BIZAPUR', 'DISTRICT_BKP CP',
'DISTRICT_BOKARO', 'DISTRICT_BOLANGIR', 'DISTRICT_BONGAIGAON',
'DISTRICT_BORDER', 'DISTRICT_BORDER DISTRICT', 'DISTRICT_BOUDH',
'DISTRICT_BUDGAM', 'DISTRICT_BULANDSHAHAR', 'DISTRICT_BULDHANA',
'DISTRICT_BUNDI', 'DISTRICT_BURDWAN', 'DISTRICT_BURHANPUR', 'DISTRICT_BUXAR',
'DISTRICT_C.I.D.', 'DISTRICT_CACHAR', 'DISTRICT_CAR', 'DISTRICT_CAW',
'DISTRICT_CBCID', 'DISTRICT_CBPURA', 'DISTRICT_CENTRAL', 'DISTRICT_CHAIBASA',
'DISTRICT_CHAMARAJNAGAR', 'DISTRICT_CHAMBA', 'DISTRICT_CHAMOLI',
'DISTRICT_CHAMPAWAT', 'DISTRICT_CHAMPHAI', 'DISTRICT_CHANDEL',
'DISTRICT_CHANDIGARH', 'DISTRICT_CHANDOLI', 'DISTRICT_CHANDRAPUR',
'DISTRICT_CHANGLANG', 'DISTRICT_CHATRA', 'DISTRICT_CHENGAI',
'DISTRICT_CHENNAI', 'DISTRICT_CHENNAI RLY.', 'DISTRICT_CHENNAISUBURBAN',
'DISTRICT_CHHATARPUR', 'DISTRICT_CHHINDWARA', 'DISTRICT_CHICKMAGALUR',
'DISTRICT_CHIRANG', 'DISTRICT_CHITRADURGA', 'DISTRICT_CHITRAKOOT DHAM',
'DISTRICT_CHITTOOR', 'DISTRICT_CHITTORGARH', 'DISTRICT_CHURACHANDPUR',
'DISTRICT_CHURU', 'DISTRICT_CID', 'DISTRICT_CID CRIME', 'DISTRICT_COIMBATORE
RURAL', 'DISTRICT_COIMBATORE URBAN', 'DISTRICT_COOCHBEHAR', 'DISTRICT_CP
AMRITSAR', 'DISTRICT_CP JALANDHAR', 'DISTRICT_CP LUDHIANA', 'DISTRICT_CRIME
BRANCH', 'DISTRICT_CRIME JAMMU', 'DISTRICT_CRIME KASHMIR', 'DISTRICT_CRIME
SRINAGAR', 'DISTRICT_CSM NAGAR', 'DISTRICT_CUDDALORE', 'DISTRICT_CUDDAPAH',
'DISTRICT_CUTTACK', 'DISTRICT_CYBERABAD', 'DISTRICT_D and N HAVELI',
'DISTRICT_DAHOD', 'DISTRICT_DAKSHIN DINAJPUR', 'DISTRICT_DAKSHIN KANNADA',
'DISTRICT_DAMAN', 'DISTRICT_DAMOH', 'DISTRICT_DANTEWARA',
'DISTRICT_DARBHANGA', 'DISTRICT_DARJEELING', 'DISTRICT_DARRANG',
'DISTRICT_DATIYA', 'DISTRICT_DAUSA', 'DISTRICT_DAVANAGERE', 'DISTRICT_DCP
BBSR', 'DISTRICT_DCP CTC', 'DISTRICT_DEHRADUN', 'DISTRICT_DELHI UT TOTAL',
'DISTRICT_DEOGARH', 'DISTRICT_DEOGHAR', 'DISTRICT_DEORIA', 'DISTRICT_DEWAS',
'DISTRICT_DHALAI', 'DISTRICT_DHAMTARI', 'DISTRICT_DHANBAD', 'DISTRICT_DHANBAD
RLY.', 'DISTRICT_DHAR', 'DISTRICT_DHARMAPURI', 'DISTRICT_DHARWAD COMMR.',
'DISTRICT_DHARWAD RURAL', 'DISTRICT_DHEMAJI', 'DISTRICT_DHENKANAL',
'DISTRICT_DHOLPUR', 'DISTRICT_DHUBRI', 'DISTRICT_DHULE', 'DISTRICT_DIBANG
VALLEY', 'DISTRICT_DIBRUGARH', 'DISTRICT_DIMAPUR', 'DISTRICT_DINDIGUL',
'DISTRICT_DINDORI', 'DISTRICT_DIU', 'DISTRICT_DODA', 'DISTRICT_DUMKA',
'DISTRICT_DUNGARPUR', 'DISTRICT_DURG', 'DISTRICT_EAST', 'DISTRICT_EAST
GODAVARI', 'DISTRICT_EOW', 'DISTRICT_ERNAKULAM', 'DISTRICT_ERNAKULAM COMMR.',
'DISTRICT_ERNAKULAM RURAL', 'DISTRICT_ERODE', 'DISTRICT_ETAH',
'DISTRICT_ETAWAH', 'DISTRICT_FAIZABAD', 'DISTRICT_FARIDABAD',
'DISTRICT_FARIDKOT', 'DISTRICT_FATEHABAD', 'DISTRICT_FATEHGARH',
'DISTRICT_FATEHGARH SAHIB', 'DISTRICT_FATEHPUR', 'DISTRICT_FAZILKA',
'DISTRICT_FEROZEPUR', 'DISTRICT_FEROZPUR', 'DISTRICT_FIROZABAD',
'DISTRICT_G.R.P', 'DISTRICT_G.R.P.', 'DISTRICT_G.R.P. AJMER',
'DISTRICT_G.R.P. JODHPUR', 'DISTRICT_G.R.P.(RLY)', 'DISTRICT_G.R.P.AJMER',
'DISTRICT_G.R.P.JODHPUR', 'DISTRICT_GADAG', 'DISTRICT_GADCHIROLI',
'DISTRICT_GAJAPATI', 'DISTRICT_GANDERBAL', 'DISTRICT_GANDHINAGAR',
'DISTRICT_GANGANAGAR', 'DISTRICT_GANJAM', 'DISTRICT_GARHWA',
'DISTRICT_GARIYABAND', 'DISTRICT_GARO HILLS EAST', 'DISTRICT_GARO HILLS

SOUTH', 'DISTRICT_GARO HILLS WEST', 'DISTRICT_GAUTAMBUDH NAGAR',
'DISTRICT_GAYA', 'DISTRICT_GHAZIABAD', 'DISTRICT_GHAZIPUR',
'DISTRICT_GIRIDIH', 'DISTRICT_GOALPARA', 'DISTRICT_GODDA',
'DISTRICT_GOLAGHAT', 'DISTRICT_GOMATI', 'DISTRICT_GONDA', 'DISTRICT_GONDIA',
'DISTRICT_GOPALGANJ', 'DISTRICT_GORAKHPUR', 'DISTRICT_GRP', 'DISTRICT_GRP
RAIPUR', 'DISTRICT_GRP(RLY)', 'DISTRICT_GULBARGA', 'DISTRICT_GUMLA',
'DISTRICT_GUNA', 'DISTRICT_GUNTAKAL RLY.', 'DISTRICT_GUNTUR',
'DISTRICT_GUNTUR URBAN', 'DISTRICT_GURDASPUR', 'DISTRICT_GURGAON',
'DISTRICT_GUWAHATI CITY', 'DISTRICT_GWALIOR', 'DISTRICT_HAILAKANDI',
'DISTRICT_HAMIRPUR', 'DISTRICT_HAMREN', 'DISTRICT_HANDWARA',
'DISTRICT_HANUMANGARH', 'DISTRICT_HARDA', 'DISTRICT_HARDOI',
'DISTRICT_HARIDWAR', 'DISTRICT_HASSAN', 'DISTRICT_HATHRAS',
'DISTRICT_HAVERI', 'DISTRICT_HAZARIBAGH', 'DISTRICT_HIMATNAGAR',
'DISTRICT_HINGOLI', 'DISTRICT_HISSAR', 'DISTRICT_HOOGHLY',
'DISTRICT_HOSHANGABAD', 'DISTRICT_HOSHIARPUR', 'DISTRICT_HOWRAH',
'DISTRICT_HOWRAH CITY', 'DISTRICT_HOWRAH G.R.P.', 'DISTRICT_HYDERABAD CITY',
'DISTRICT_I.G.I. AIRPORT', 'DISTRICT_IDUKKI', 'DISTRICT_IGI AIRPORT',
'DISTRICT_IMPHAL EAST', 'DISTRICT_IMPHAL WEST', 'DISTRICT_IMPHAL(EAST)',
'DISTRICT_IMPHAL(WEST)', 'DISTRICT_INDORE', 'DISTRICT_INDORE RLY.',
'DISTRICT_J.P.NAGAR', 'DISTRICT_JABALPUR', 'DISTRICT_JABALPUR RLY.',
'DISTRICT_JAGATSINGHPUR', 'DISTRICT_JAGDALPUR', 'DISTRICT_JAGRAON',
'DISTRICT_JAINTIA HILLS', 'DISTRICT_JAIPUR', 'DISTRICT_JAIPUR EAST',
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SPITI', 'DISTRICT_LAKHIMPUR', 'DISTRICT_LAKHISARAI', 'DISTRICT_LAKSHADWEEP',
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```python
import pandas as pd
from sklearn.feature_selection import SelectKBest, chi2

# ==============================
# Load Dataset
# ==============================
file_path = r"C:\Users\Dell\Documents\Modified_Crime_Dataset.csv"
data = pd.read_csv(file_path)

# Recreate 'Total_Crimes' column if it doesn't exist
if 'Total_Crimes' not in data.columns:
    crime_columns = data.columns[3:]  # Adjust based on your dataset
    data['Total_Crimes'] = data[crime_columns].sum(axis=1)

# Define features (X) and target variable (y)
X = data.drop(columns=['Total_Crimes', 'STATE/UT', 'DISTRICT', 'YEAR'],
        errors='ignore')  # Drop non-numeric columns
y = data['Total_Crimes']

# ==============================
# Chi-Square Test
# ==============================
# Discretize the target variable into 4 bins
y_binned = pd.qcut(y, 4, labels=False, duplicates='drop')

# Discretize numeric features
def discretize_column(col):
    if len(col.unique()) > 4:  # Only discretize if there are enough unique
        values
        return pd.qcut(col, 4, labels=False, duplicates='drop')
    return col

X_discretized = X.apply(discretize_column)

# Select top 4 features using the chi-squared test
chi_selector = SelectKBest(chi2, k=4)
```

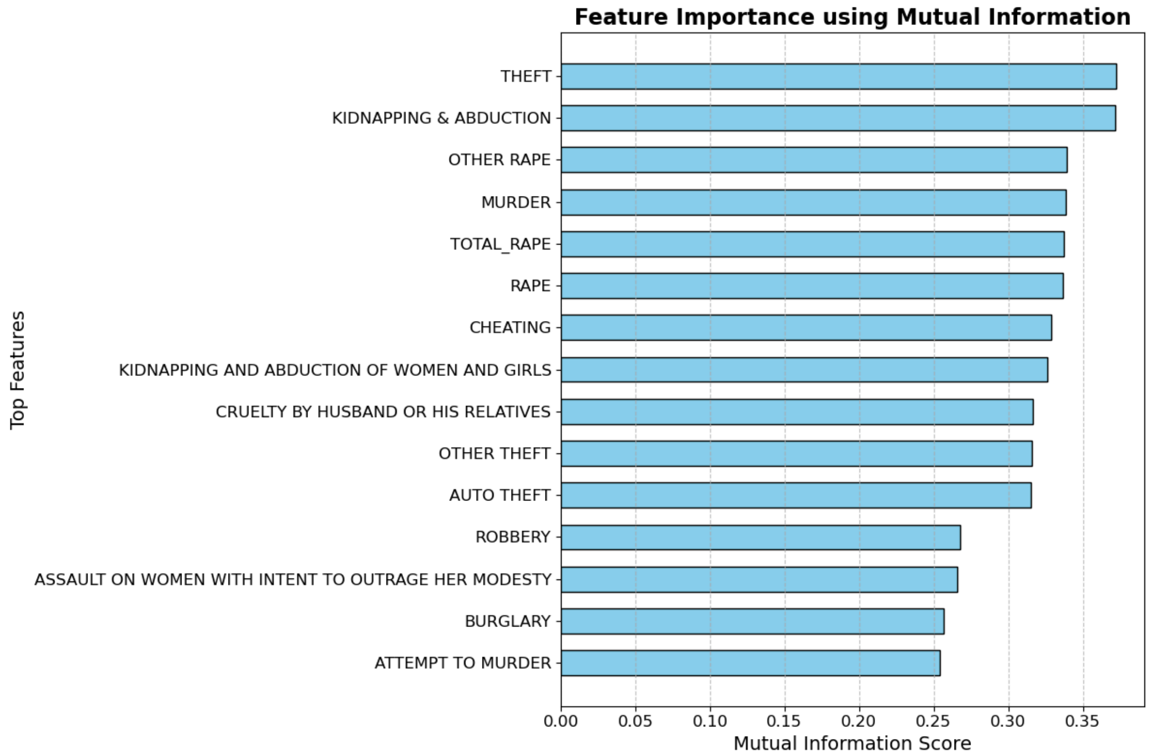
```python
X_chi2 = chi_selector.fit_transform(X_discretized, y_binned)

# Get selected feature names
selected_features_chi2 =
         X_discretized.columns[chi_selector.get_support()].tolist()
print("Selected Features using Chi-Square Test:", selected_features_chi2)
```

```
Selected Features using Chi-Square Test: ['RAPE', 'OTHER RAPE', 'THEFT',
'TOTAL_RAPE']
```

```python
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.preprocessing import MinMaxScaler
from sklearn.feature_selection import mutual_info_regression

# ==============================
# Load Dataset
# ==============================
file_path = r"C:\Users\Dell\Documents\Modified_Crime_Dataset.csv"
data = pd.read_csv(file_path)

# Filter the dataset to keep only state-level data
# Drop 'DISTRICT' column if it exists
data = data.drop(columns=['DISTRICT'], errors='ignore')

# Recreate 'Total_Crimes' column if it doesn't exist
if 'Total_Crimes' not in data.columns:
    crime_columns = data.select_dtypes(include=['int64', 'float64']).columns
    data['Total_Crimes'] = data[crime_columns].sum(axis=1)

# Define features (X) and target variable (y)
# Drop columns that are not meaningful for feature selection
X = data.drop(columns=['Total_Crimes', 'STATE/UT', 'YEAR'], errors='ignore')
         # Keep numeric features only
y = data['Total_Crimes']

# Ensure all features are numeric
X = X.select_dtypes(include=['int64', 'float64'])

# ==============================
# Mutual Information Calculation
# ==============================
# Scale the features to [0, 1] for MI calculation
scaler = MinMaxScaler()
X_scaled = scaler.fit_transform(X)

# Calculate mutual information scores
mi_scores = mutual_info_regression(X_scaled, y, random_state=42)
mi_df = pd.DataFrame({'Feature': X.columns, 'MI Score': mi_scores})
mi_df = mi_df.sort_values(by='MI Score', ascending=False)

print("\nFeature Importance using Mutual Information:")
print(mi_df)
```

```python
# ==============================
# Improved Visualization
# ==============================
# Filter top 15 features for better readability
top_features = mi_df.head(15)

plt.figure(figsize=(12, 8))  # Increase figure size for better clarity
plt.barh(top_features['Feature'], top_features['MI Score'], color='skyblue',
         edgecolor='black', height=0.6)  # Tighter bars
plt.xlabel('Mutual Information Score', fontsize=14)
plt.ylabel('Top Features', fontsize=14)
plt.title('Feature Importance using Mutual Information', fontsize=16,
         fontweight='bold')
plt.xticks(fontsize=12)
plt.yticks(fontsize=12)
plt.grid(axis='x', linestyle='--', alpha=0.7)  # Add light grid lines for x-
         axis
plt.gca().invert_yaxis()  # Invert y-axis for better readability
plt.tight_layout()  # Adjust layout to avoid clipping
plt.show()
```

```
Feature Importance using Mutual Information:
                                               Feature  MI Score
13                                               THEFT  0.372469
6                               KIDNAPPING & ABDUCTION  0.371510
5                                           OTHER RAPE  0.339027
0                                               MURDER  0.338636
28                                          TOTAL_RAPE  0.337268
3                                                 RAPE  0.336831
18                                            CHEATING  0.328957
7         KIDNAPPING AND ABDUCTION OF WOMEN AND GIRLS  0.326396
25                CRUELTY BY HUSBAND OR HIS RELATIVES  0.316292
15                                         OTHER THEFT  0.315552
14                                          AUTO THEFT  0.315160
11                                             ROBBERY  0.267658
23  ASSAULT ON WOMEN WITH INTENT TO OUTRAGE HER MO...  0.265992
12                                            BURGLARY  0.256734
1                                    ATTEMPT TO MURDER  0.254200
21                                  HURT/GREVIOUS HURT  0.245273
17                            CRIMINAL BREACH OF TRUST  0.232257
27                         CAUSING DEATH BY NEGLIGENCE  0.217278
22                                        DOWRY DEATHS  0.195566
16                                               RIOTS  0.193676
8                   KIDNAPPING AND ABDUCTION OF OTHERS  0.183835
20                                               ARSON  0.154947
19                                      COUNTERFEITING  0.150308
2          CULPABLE HOMICIDE NOT AMOUNTING TO MURDER  0.139809
9                                              DACOITY  0.139200
10               PREPARATION AND ASSEMBLY FOR DACOITY  0.099344
24                        INSULT TO MODESTY OF WOMEN  0.096390
26         IMPORTATION OF GIRLS FROM FOREIGN COUNTRIES  0.014238
4                                       CUSTODIAL RAPE  0.004786
```




```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Compute the correlation matrix
correlation_matrix = pd.concat([X, y], axis=1).corr()

# Filter top features to focus on the most relevant relationships (optional)
top_features = correlation_matrix.abs().nlargest(10, 'Total_Crimes')
         ['Total_Crimes'].index
filtered_correlation_matrix = correlation_matrix.loc[top_features,
         top_features]

# Set up the matplotlib figure
plt.figure(figsize=(10, 8))  # Adjust figure size for better clarity

# Draw the heatmap
sns.heatmap(
    filtered_correlation_matrix,
    cmap='coolwarm',  # Use a clean color palette
    annot=True,  # Annotate with correlation values
    fmt=".2f",  # Limit decimals for neatness
    linewidths=0.5,  # Add thin lines between boxes
    square=True,  # Keep cells square for consistency
    cbar_kws={"shrink": 0.8},  # Shrink the color bar
    annot_kws={"size": 10}  # Adjust font size for annotations
)

# Set the title and axis labels
```

```python
plt.title('Correlation Matrix of Key Features', fontsize=14,
         fontweight='bold')
plt.xticks(fontsize=10, rotation=45, ha='right')  # Rotate x-axis labels
plt.yticks(fontsize=10)  # Adjust y-axis label size
plt.tight_layout()  # Ensure everything fits neatly
plt.show()
```